

Headache COAR

Objectives

Clinical Skills

1. Identify important aspects of a HPI relevant to the patient who presents with a headache
2. Identify components of a physical exam of a patient with a headache, including HEENT, neurological, cardiovascular, and musculoskeletal considerations
3. Review the cranial nerve exam
4. Create and narrow down a differential diagnosis for headache
5. Create an osteopathic treatment plan for the patient with a headache
6. Describe pharmacologic therapy for the patient with headache
7. Describe red flags related to headaches

Osteopathic Manipulative Medicine

1. Integrate an osteopathic structural exam into your physical exam
2. Discuss the A (Autonomics), B (Biomechanics), C (Circulation), and S (Screening) aspects of how structural findings could contribute to headache
3. Perform osteopathic manipulative treatment aimed to assist a patient who presents with a headache
4. Identify how viscerosomatic reflexes may contribute to headaches

Anatomy

1. Review innervation/nociceptive patterns of the head and cranial dura and their anatomic relationships, including: Cervical plexus, Greater Occipital N., Lesser Occipital N., Trigeminal N., Vagus N.
2. Discuss the muscles, bones, joints, fascial systems of the upper cervical spine and head and their potential involvement in headache pathology
3. Describe the role of the trigeminocervical ganglion in headache.

Radiology

1. Read a normal head CT and identify major cranial structures including foramina and their contents.
2. Determine proper imaging studies warranted for a patient with a headache

History

Location: Emergency Department

CC: Headache

HPI: 22-year-old Hispanic female with PMHx of obesity presents to the Emergency Department with a “really bad headache”.

Patient reports that approximately 4 hours ago she was pumping gas in her car when she noticed visual changes, including bright geometrical shapes. This progressed to partial loss of vision with continued bright spots. She was worried, and so instead of going to work she went home. The visual changes continued, followed by tingling and ascending numbness in the right UE (started in the fingers). Numbness spread to her face, especially perioral region, lips, and tongue. She then attempted to call the advice nurse but was having difficulty with speech and ringing in her ears, so a family member drove her to the hospital.

Over the course of the following hour, she developed very severe head pain (9/10) with associated +nausea, +vomiting x 3, and +photophobia. In the ER, the patient reports the visual “light show” and arm and face numbness had stopped, but now her whole head is throbbing, worse on the right. Noise, lights, and movement make the headache worse; nothing makes it better although she has not tried any medication yet. She reports some radiation of the pain and muscle tightness in the neck and upper back region. Patient denies fevers, chills, hearing loss, chest pain, dyspnea, palpitations, bloody emesis, abdominal pain, changes in bowel or bladder function, muscle weakness. She has never had a headache like this before. She notes no significant recent trauma, accident or injury, though she did “back-flop” into a pool while on vacation 2 months ago. She had an immediate headache afterwards that lasted a few days with increased fatigue then resolved.

PMH: Obesity, hospitalized for pneumonia at age 12

PSH: laparoscopic cholecystectomy age 19, wisdom teeth (4) removal age 17

Trauma history: Fall “back flop” on water, +concussion at age 22 (2 months prior to presentation), MVA-rear-ended restrained driver at age 19 with whiplash injury

Allergies: NKDA, NKFA, no latex allergies

Medications: Combined estrogen-progesterone oral contraception

Social History: admits to smoking ¼ a pack per day since age 18, denies illicit drug use, occasional/social EtOH; single, sexually active in a monogamous relationship for 1 year with a male partner; works in a call center, lives with parents, no pets. Life stressors- grandfather whom she was close with recently passed away. Work is new and somewhat stressful.

Family History: Mother-alive 48: healthy; Father-alive 50: obesity, osteoarthritis; sister-24: obesity, PCOS

Patient Concerns: “I am worried I am having a stroke; this is really scary”

ROS:

General: denies fevers, chills, night sweats, fatigue, or weight changes

HEENT: denies sore throat, rhinorrhea, congestion, otalgia, dysphagia, sinusitis, vertigo, jaw pain, bruxism, seasonal allergies. +photophobia, +phonophobia, +tinnitus, + wears contacts- last eye exam 1 year ago, + visual sx's as per HPI

Resp: denies cough, shortness of breath

Cardiac: denies chest pain, palpitations

GI: denies abdominal pain, hematemesis, diarrhea, constipation. +nausea, +vomiting

Neuro: denies muscle weakness, seizure, loss of consciousness, ataxia. +numbness, +tingling, +vision changes

MSK: admits to head, neck, and upper back pain as per HPI. Denies other arthralgias, myalgias, joint swelling or joint redness.

Derm: denies rash or skin changes

Endo: denies polyuria, polydipsia, changes in hair, skin or nails, heat or cold intolerance

Heme: denies easy bleeding or bruising

Physical Exam

Vitals: HR: 88, RR 14, BP 110/79, BMI 31, Temp 98.0, pO₂ 98 on RA

General: well-developed well-nourished (WD/WN) female in moderate distress lying on the table lateral recumbent with hood covering eyes and green emesis in bowl on side table.

HEENT: NC/AT, EOMI, PERRLA, anicteric, no injection, no papilledema, fundus WNL, bilateral EAM normal, and TMs intact with good cone of light with no injection or effusion; nares clear and patent without injection, nasal septum midline; oropharynx clear with no erythema or exudate, good dentition, TMJ with normal motion without clicks or deviation, no sinus pressure tenderness

Neck: no LAD (lymphadenopathy), thyroid without goiter or nodules, neck supple

Lungs: CTAB (clear to auscultation bilaterally), no R/R/W (rales/rhonchi/wheezes)

Cardiovascular: RRR (regular rate and rhythm), no M/R/G (murmurs/rubs/gallops), normal S1/S2, PMI nondisplaced, +2/4 pulses at carotid, radial, dorsalis pedis

ABDOMEN: normoactive BS (bowel sounds) x 4, soft, NT/ND (non-tender, non-distended), no organomegaly, well-healed surgical scars noted. Aorta normal size. No bruit of the aorta or renal vessels.

Neuro: CN II- XII intact to direct testing without focal deficits, UE DTR +2/4, LE DTR +2/4, normal gait, UE/LE strength +5/5 equal and symmetric, down going Babinski, negative Hoffman's, no Kernig's or Brudzinski's sign, no dysarthria, peripheral vision intact, sharp-dull sensation intact all 4 extremities

MSK: cervical spine with decreased ROM in left rotation and sidebending, TART changes at OA, AA with bilateral hypertonic paraspinal musculature

Skin: no rash

Osteopathic Structural Exam:

Head: OA ER_RSL, right OM suture restricted, right temporal IR (internally rotated), decreased motion on the right side of the cranium and fullness on the left, left sidebending rotation at SBS, decreased CRI

Cervical: AC1, AC2, AC3 counterstrain points, C3 ER_RSR, C4 ER_LSL, C5 ER_RSR, AA R_R

Thoracic: T1 FR_RSR, T2-T4 NS_RRL, T5 ER_RSR, hypertonic and tender superior trapezius and levator scapulae muscles bilaterally

Lumbar: L/S junction rotated right, L1-L3 NS_RRL

Sacrum: L on L sacral torsion, decreased motion with respiration

Pelvis: right innominate anteriorly rotated, right innominate inflare, right inferior pubic shear

Ribs/Sternum: right ribs 1-2 inhaled

Abdomen: Diaphragm rotated left, sidebent left

Zink: OA: R; C/T: R (most significant restriction); T/L: R; L/S: R

Clinical Skills

Everyone performs their integrated physical exam slightly differently. Some of you may wish to separate the HEENT exam from the CN exam, while others will choose to integrate it together as noted below. While there is no correct answer, it is important to practice and develop your routine because if you perform your physical exam the same way each time then you are less likely to miss things!

Anterior Seated

Head and Neck

- Observe: General posture (including shoulders) and head carriage, facial symmetry
- Inspect: scalp, skull, and hair for symmetry, trauma, or lesions; AROM of cervical spine
- Palpate: face, suboccipital muscles and neck for muscle hypertonicity, lymphadenopathy, trachea, Chapman's points, sinus pressure tenderness; palpation of supraorbital, infraorbital, and mental foramen
- TMJ palpation, jaw opening/closing, palpation of temporal region and mastoid
- CN Exam: Facial sensation/jaw movements/corneal reflexes (CN V), facial symmetry and strength (CN VII)

Eyes

- Observe: symmetry, pupil size, erythema
- Inspect: conjunctiva and sclera
- Pupillary reaction and accommodation (direct and consensual response) (CN II/III)
- Fundoscopic exam for papilledema
- Visual acuity and visual field testing (CN II)
- CN Exam: EOM (CN III, IV, VI), check for nystagmus

Ears

- Observe: ear position
- Inspect: external ear and external auditory canal
- Otoscopic Exam: auditory canal and tympanic membranes
- Palpate: mastoid process, temporal region
- Auditory acuity (whispered voice test)
- CN Exam: hearing (CN VIII)

Nose

- Observe: symmetry, nasal septum, nasal mucosa
- Inspect: nares with otoscope
- CN Exam: smell (CN I)

Throat (Mouth and Pharynx)

- Observe: lips and muscular folds around the mouth, oral mucosa
- Inspect: tongue, palate, oropharynx, dentition, tonsils
- Palpate: temporomandibular joint for clicks or deviation (if not already done above)
- CN Exam: bite (CN V), tongue symmetry/position/movement (CN XII), uvula deviation (CN X), swallowing and rise of the palate/gag reflex (CN IX, X)

Neuro

- Observe: gait (tandem walk, heel/toe), speech
- Inspect: muscle strength, muscle bulk, muscle tone, coordination, rapid alternating movement, sensation, proprioception, standing flexion test
- Deep tendon reflexes: upper and lower extremities
- Plantar response (Babinski), , Pronator Drift, Hoffman reflex
- CN Exam: voice and speech (CN V, VII, X, XII), shoulder and neck movements (CN XI)

Posterior Seated

- Observe: position of the thorax and cervical spine, cervical AROM
- Palpate: thyroid, swallow, cervical PROM, paraspinals, scalenes, SCM, upper trapezius, Zink Screen- O/A, C/T, T/L, L/S junctions, thoracic spine (short lever) and TART
- Perform: auditory acuity

Standing

- Observe gait (tandem walk, walking on heels, walking on toes) general posture
- Observe: standing flexion test if time permits
- Perform: Romberg test

Supine

- Observe: neck mobility, nuchal rigidity (Brudzinski sign, Kernig sign)
- Palpate: cranial motion- vault contact, static symmetry, dynamic evaluation, SBS motion, temporal bone motion, parietal bone motion, occipitomastoid suture, anterior and posterior cervical counterstrain points, sacral motion

Differential Diagnosis

Below is a list of potential diagnoses based on the present case. How would you rule in/out some of these differential diagnoses? Use UpToDate or another peer reviewed source to look up each diagnosis and start to create a comparison grid, to understand overlapping symptoms, presentations, physical exam, etc.

- Cerebral aneurysm
- Stroke/cerebrovascular accident (CVA) or transient ischemic attack (TIA)
- Dissection: internal carotid artery, vertebral artery
- Hemorrhage: epidural, subdural, or subarachnoid
- Chiari malformation
- Infections: meningitis, encephalitis (herpes simplex)
- Space-occupying lesion (tumor, abscess)
- Pseudotumor cerebri
- Vasculitis, such as temporal arteritis
- Cluster headache
- Migraine Headache (variants- common, classic, hemiplegic, retinal, chronic)

- Occipital neuralgia
- Paroxysmal hemicrania
- Tension headache
- Nummular headache
- TMJ dysfunction
- Sinusitis
- Tolosa-Hunt Syndrome
- Trigeminal Neuralgia
- Acute glaucoma
- Concussion (mTBI) or post-concussion syndrome
- Psychiatric disease (somatization)
- Rebound headache
- Illicit drug use or prescription medication
- Whiplash injury, cervical strain
- Somatic dysfunction
- Cervicogenic headache
- Eye strain
- Optic neuritis
- Pheochromocytoma
- Carbon monoxide poisoning

Anatomy

What causes the pain of a headache? The brain itself does not have nociceptive fibers, so when there is a headache, it is due to nociception of other structures in the head with nociceptive input.

- **Extracranially:**
 - Scalp
 - Arteries
 - Muscle
 - Mucous membranes
 - External auditory meatus
 - Teeth
- **Intracranially:**
 - Meninges
 - Venous sinuses
 - Arteries
- **Other structures** innervated by C1, C2, C3 can also contribute to headache pain either through innervation or referral patterns
 - Suboccipital muscles
 - Muscles of the neck
 - Zygapophysial joints of C2-C4

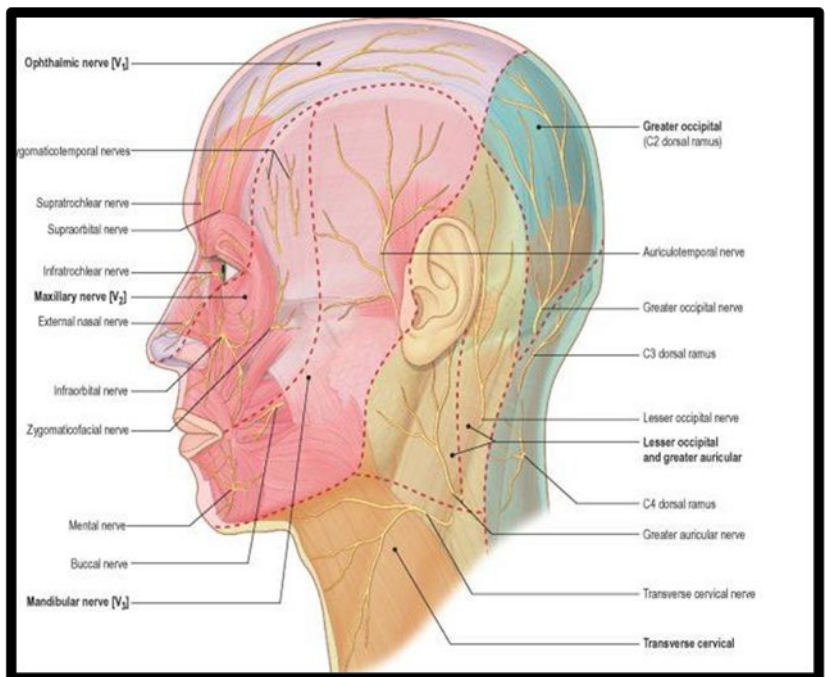
- OA (C1) and AA (C2)
- Dura of the spinal cord, posterior fossa, and inferior surface of the tentorium cerebelli
- Vertebral artery
- Trapezius and SCM

Innervation considerations:

Ventral and Dorsal Cervical Plexus

The ventral cervical plexus arises from spinal nerves C1-C4 and provides sensory innervation to the neck and head as well as motor innervation to the musculature of the anterior neck. The sensory component of this plexus will exit posterior to the sternocleidomastoid muscle and split into 4 major named nerves and dural branches:

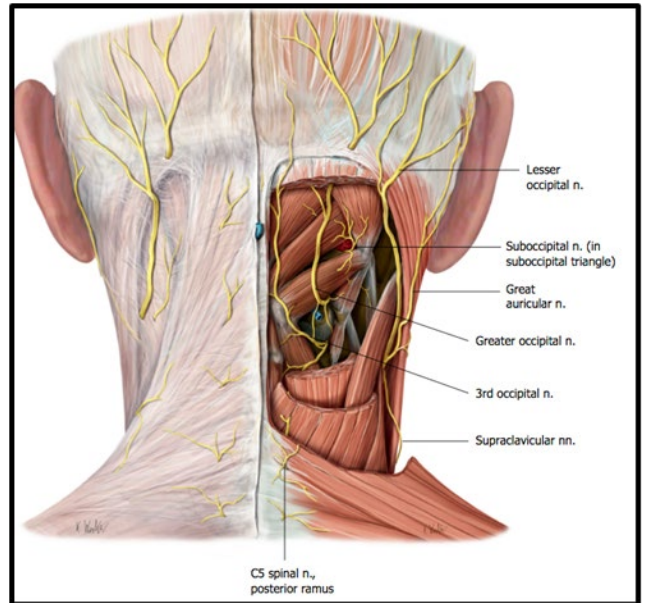
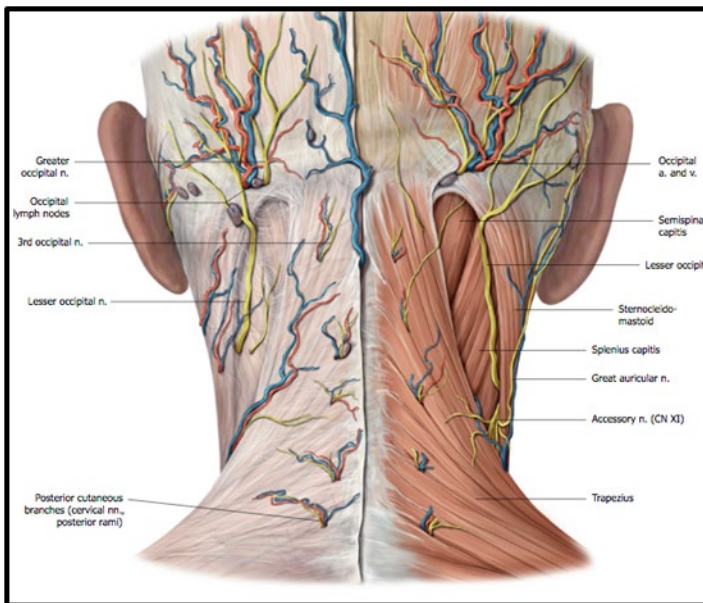
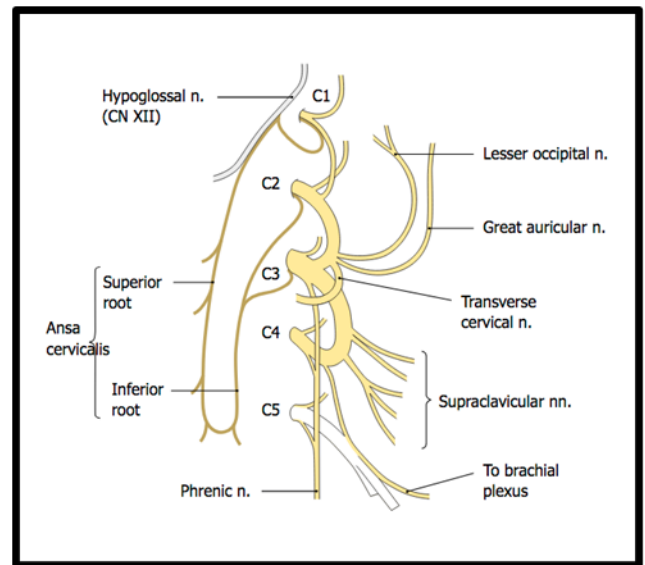
- **Lesser occipital n. C2** - Sensory to skin of the lateral head posterior to the ear
- **Great auricular n. C2, C3** - Sensory to the inferior ear, skin over the parotid gland and mastoid process
- **Transverse cervical n. C2, C3** - Sensory to the entirety of the anterior neck (passes deep to the platysma and sends sensory branches through the muscle).
- **Supraclavicular n. C3, C4** - Divides into medial, intermediate and lateral branches to supply sensory innervation over the clavicle, pectoralis major, deltoid and parts of the posterior shoulder.
- **Dural branches** - Nerves from the C2, C3 spinal nerves will ascend through the foramen magnum, but also travel with the vagus nerve and hypoglossal nerve to innervate the dura mater of the posterior cranial fossa.



Motor innervation is transmitted through the ansa cervicalis (originates from superior (C1-C2) and inferior (C2-C3) roots of cervical plexus) and will control the infrahyoid muscles (excluding thyrohyoid). Additionally, fibers from the ventral cervical rami will form the phrenic nerve (C3-C5) and will innervate the anterior, middle and posterior scalenes (C4-C6), longus colli/capitis, and along with CN X1, the trapezius and sternocleidomastoid muscles.

The dorsal cervical plexus (Plexus of Cruveilhier) receives contributions from the dorsal rami of C1-C3 and will distally form its own nerve. Like the ventral plexus, all three nerves have connecting fibers with each other. In addition to the following, the dorsal rami of the dorsal cervical plexus innervates the surrounding ligaments of the cervical spine, the facet (zygapophyseal) joints segmentally and the spinal dura.

- **Suboccipital n. C1** - Motor innervation to suboccipital muscles: rectus capitis posterior major, rectus capitis posterior minor, obliquus capitis superior and obliquus capitis inferior. C1 has no cutaneous nerve branches.
- **Greater occipital n. C2** - Sensory to the posterior head. Motor to posterior neck musculature: splenius capitis, semispinalis capitis. The greater occipital nerve passes directly through the semispinalis capitis muscle.
- **3rd occipital n. C3** - Provides sensory innervation to the posterior neck and lower part of the head.

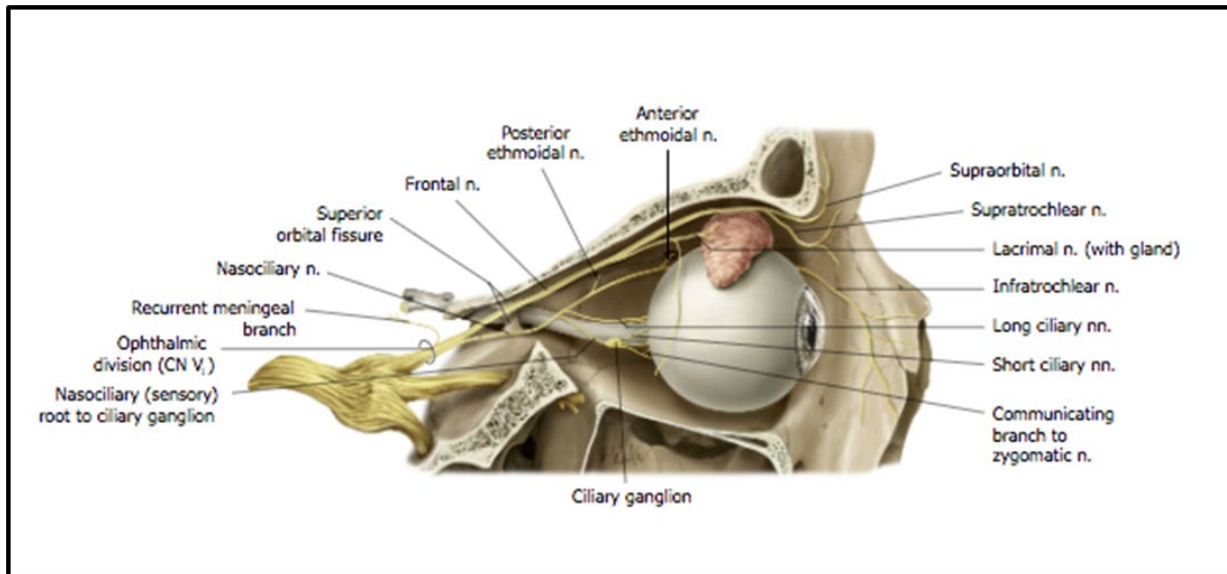


Trigeminal Nerve - CN V

The trigeminal nerve is the most widespread cranial nerve in the head and neck, providing sensory innervation to the face as well as motor innervation to the muscles of mastication and serves as a conduit for sympathetic neurons to the head. Three large areas of the face can be mapped out as peripheral nerve regions indicated by the three divisions of the trigeminal nerve: Ophthalmic V1, Maxillary V2 and Mandibular V3.

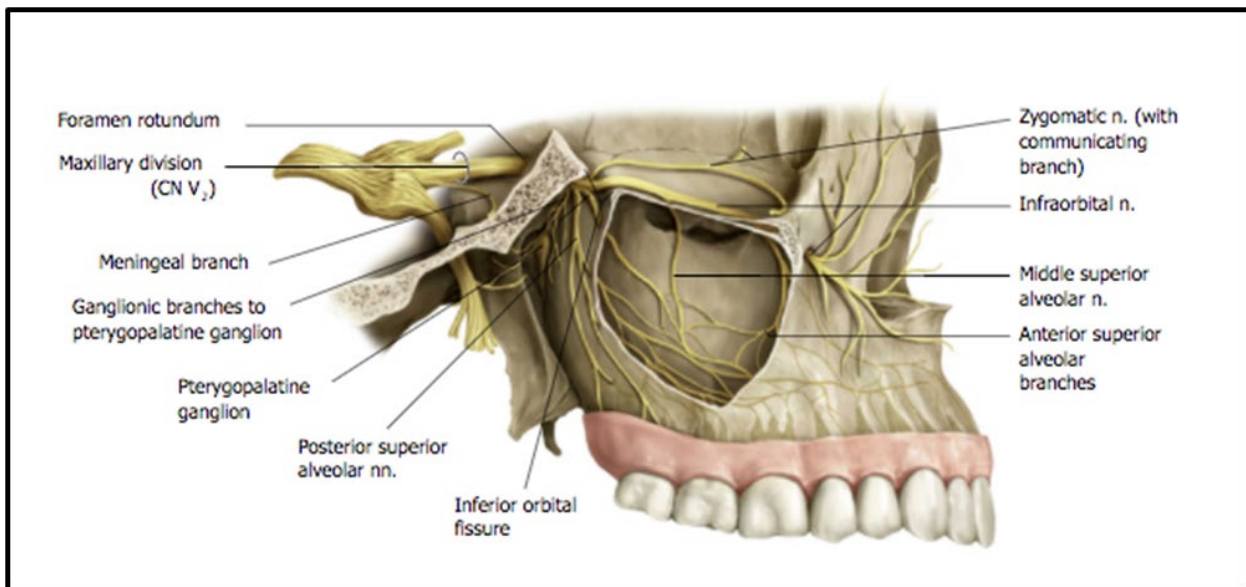
Ophthalmic Nerve V1 - Sensory only - The ophthalmic nerve exits the cranium through the superior orbital fissure and divides into 3 main branches:

- **Lacrimal n.** - Sensory to the lacrimal gland, conjunctiva, and upper eyelids. Carries parasympathetic innervation to the lacrimal gland and mucous membranes.
- **Frontal n.** - Divides into supraorbital and supratrochlear nerves to provide sensory innervation to the forehead and scalp
- **Nasociliary n.** - Sensory to the proximal part of the nose. The nasociliary nerve contains autonomic fibers for the mucous membranes and eye.
- **Dural branches** - Sensory to anterior dura and supratentorial dura mater



Maxillary nerve V2 - Sensory only - The maxillary division exits the cranium through the foramen rotundum of the sphenoid bone and divides into many branches. Only a few will be discussed here.

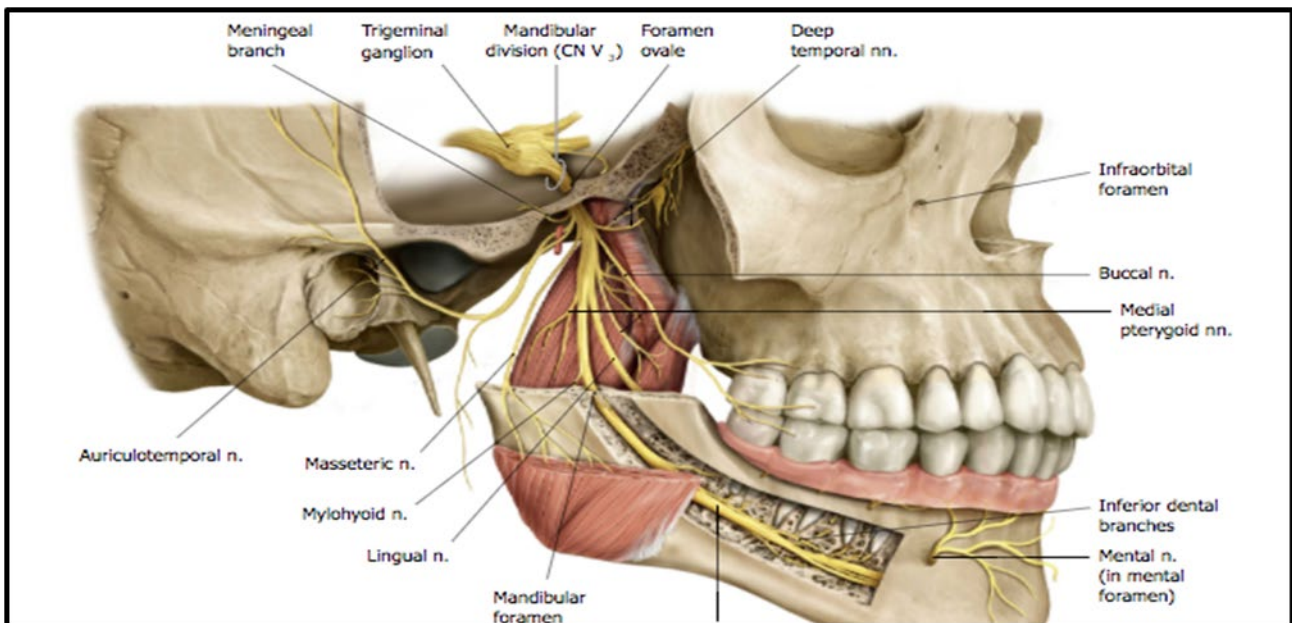
- **Zygomatic n.** - Sensory innervation to the temple and cheek, forms an anastomosis with the zygomatic branch of the facial nerve CN VII
- **Infraorbital n.** - Exits the infraorbital foramen of the maxillary bone and provides sensory innervation to the nose, cheek and upper lip. The infraorbital nerve also forms an anastomosis with facial nerve branches.



- **Superior alveolar n.** - Sensory to the maxillary sinus, palate, and maxillary teeth.
- **Dural branches** - Sensory innervation to the dura in the anterior and middle cranial fossa.

Mandibular nerve V3 - Sensory and motor - The mandibular division of the trigeminal nerve exits the skull through the foramen ovale of the greater wing of the sphenoid bone and immediately divides into 4 major branches. The mandibular division is associated with two parasympathetic ganglia: otic and submandibular.

- **Buccal n.** - Sensory innervation to the skin over the buccinator muscle. Forms an anastomosis with the buccal branches of the facial nerve.
- **Lingual n.** - Sensory to the anterior two-thirds of the tongue and floor of the mouth. Carries taste to the anterior two-thirds of the tongue via the chordae tympani of the facial nerve.
- **Auriculotemporal n.** - Emerges onto the face near the temporomandibular joint and provides sensory innervation to the joint and skin around the tragus of the ear, auricle, and the temporal region. Forms an anastomosis with the zygomatic and temporal branches of the facial nerve.
- **Inferior alveolar n.** - Travels over the lateral pterygoid muscle and gives off the nerve to mylohyoid, which innervates the mylohyoid and anterior digastric muscles before entering the mandibular foramen. Upon entering the mandible, the nerve provides sensory innervation to the mandibular teeth before exiting the mandible through the mental foramen as the mental nerve. The mental nerve provides cutaneous innervation to the chin and lower lip.
- **Dural branches** - Sensory innervation to the dura of the middle and inferior cranial fossa.

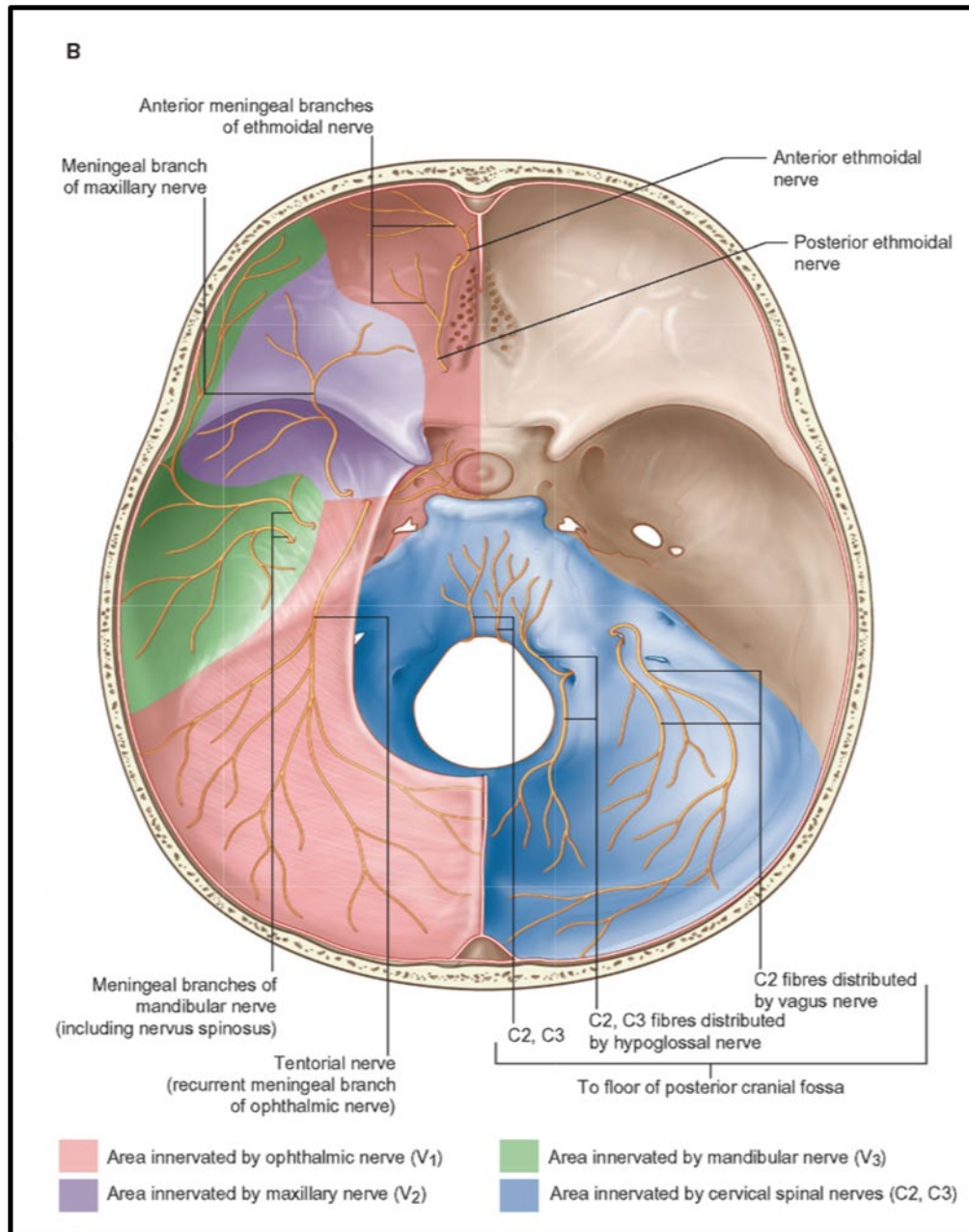


The motor division of the trigeminal nerve will provide innervation to the muscles of mastication: temporalis, masseter, lateral and medial pterygoid muscles. Additionally, V3 will innervate the tensor tympani, tensor veli palatini, mylohyoid, and the anterior digastric muscles.

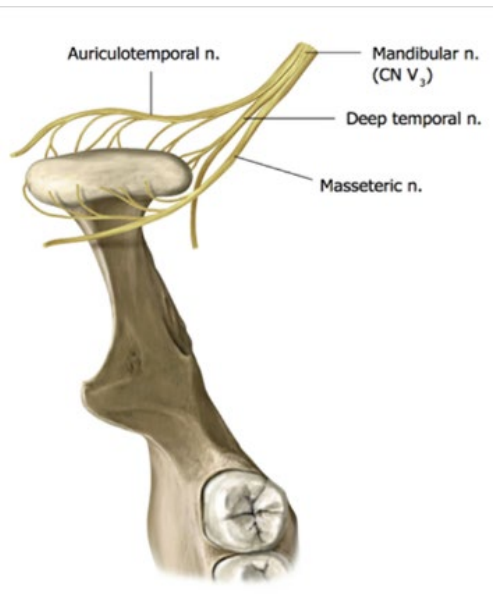
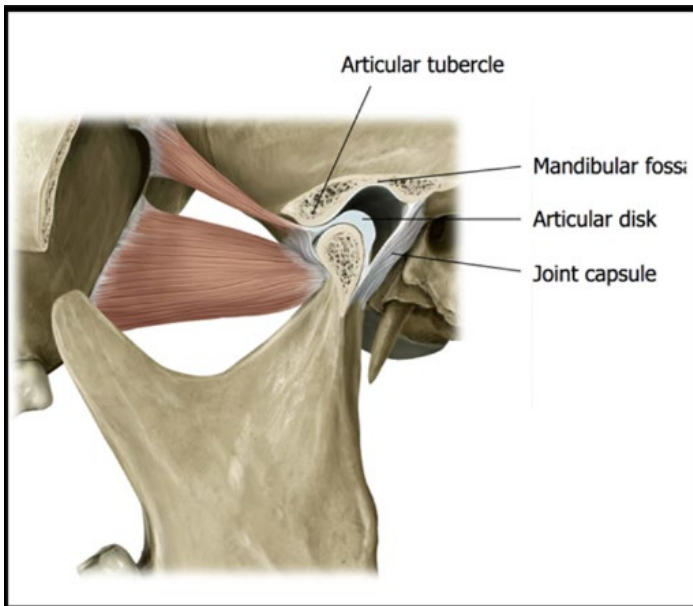
Vagus Nerve - CN X

The vagus nerve is one of the most extensive nerves in the body providing parasympathetic innervation to the thorax and abdomen; but for our purposes here we will only discuss its distribution in the head. The vagus nerve, after passing through the jugular foramen, will give off two branches to the cranium.

- **Meningeal/dural branches** - Provides sensory innervation to the dura mater of the posterior cranial fossa.
- **Auricular branch** - Also known as Arnold's nerve, this branch joins with a nerve from the glossopharyngeal nerve (CN IX) and provides an ascending branch to the facial nerve. The auricular branch supplies sensory innervation to the skin of part of the ear and the external auditory meatus.



Temporomandibular Joint



- Innervation of the TMJ is from the mandibular branch of the trigeminal nerve.

Arterial Blood Flow in the Neck and Cranium

Blood from the heart is transported to the head and neck via bilateral carotid and vertebral arteries. The carotid will bifurcate into the external and internal carotid arteries round the C3-C4 vertebral level. The internal will terminate at the Circle of Willis to supply the brain and the external will supply the neck and face. You may remember the mnemonic for the external branches as SALFOPMS.

Superior thyroid a.

Ascending pharyngeal a.

Lingual a.

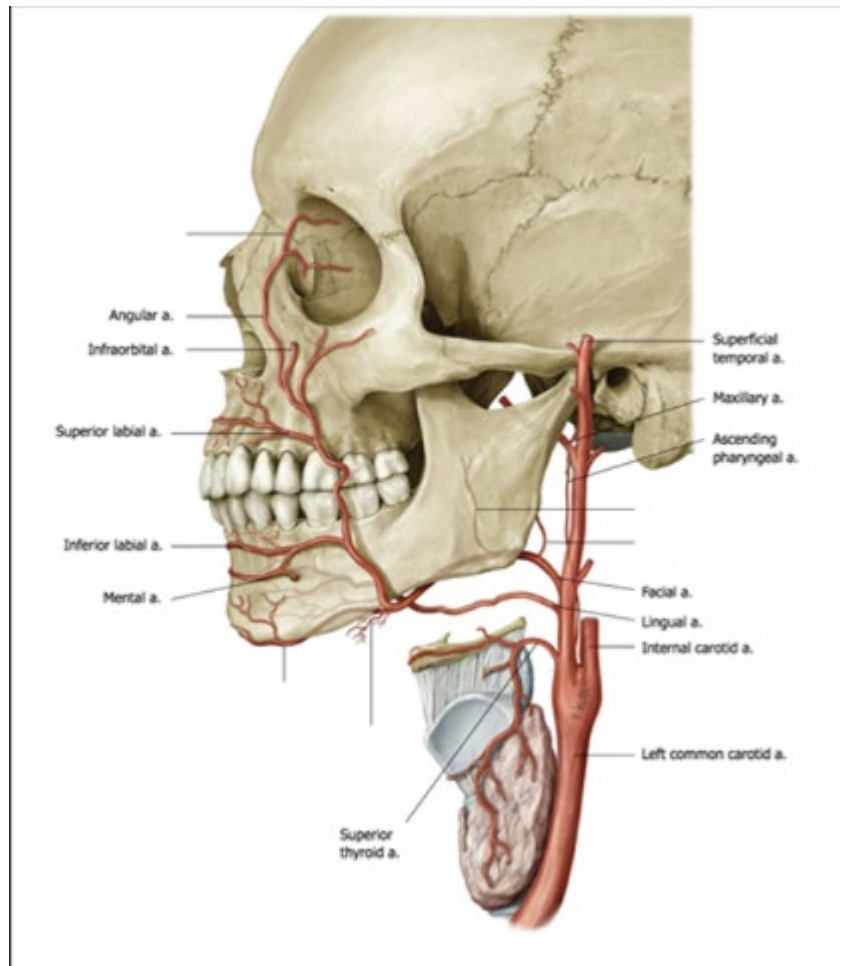
Facial a.

Occipital a.

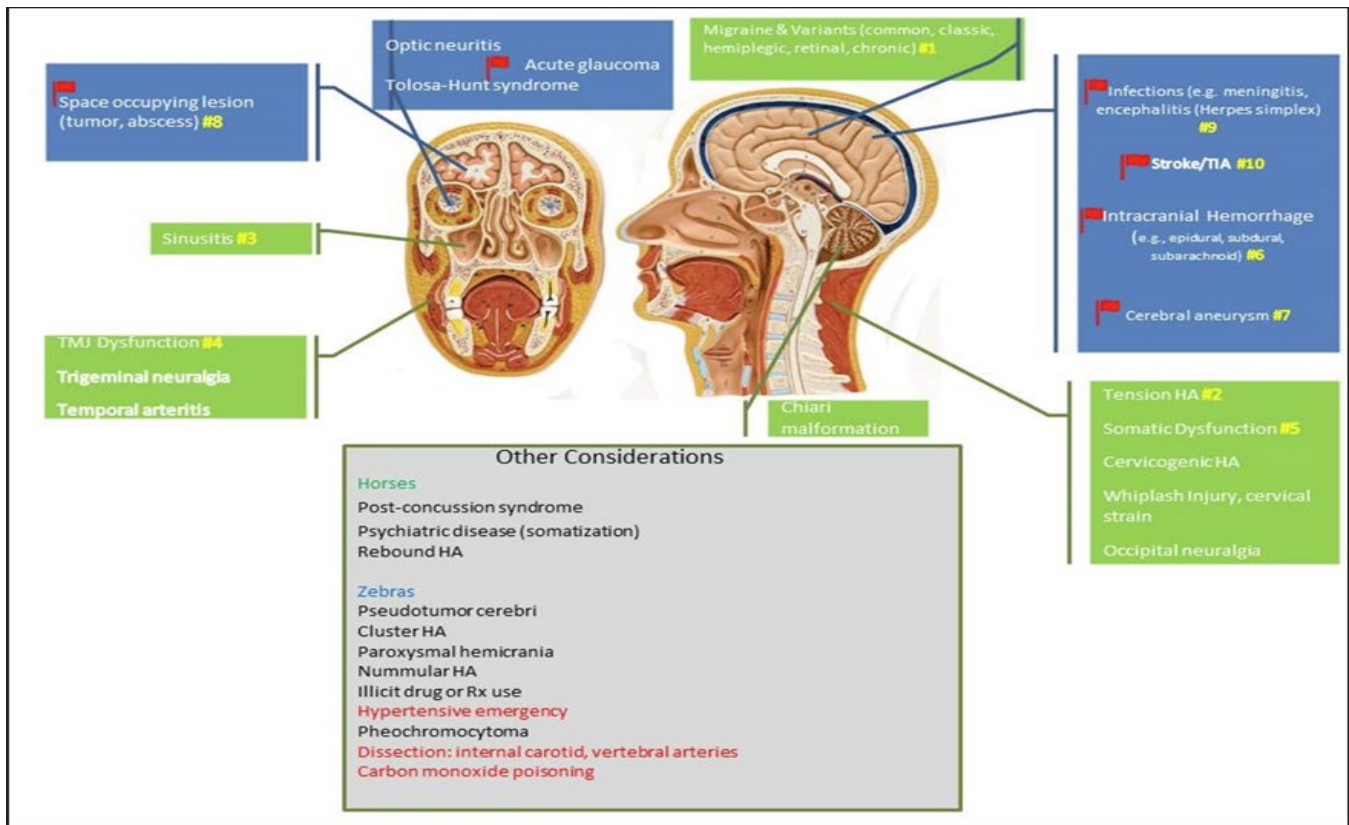
Posterior auricular a.

Maxillary a.

Superficial temporal a.



Assessment



Our patient presented with:

Pertinent Positives:

- Very severe headache
- Headache onset over course of hours
- Visual changes typical of aura
- Transient paresthesia in RUE and face
- Transient expressive aphasia
- Tinnitus
- Unilateral head pain (R side)
- Photophobia
- Phonophobia
- Neck and upper back tightness
- Nausea/vomiting
- History of head trauma/concussion 2 months prior to presentation
- History of whiplash injury 3 years prior to presentation
- Wears contact lenses with fairly recent eye exam
- Meds: OCPs
- Social: +tobacco
- Stressors: grief due to passage of grandfather recently
- Current anxiety
- Somatic dysfunction in cervical region

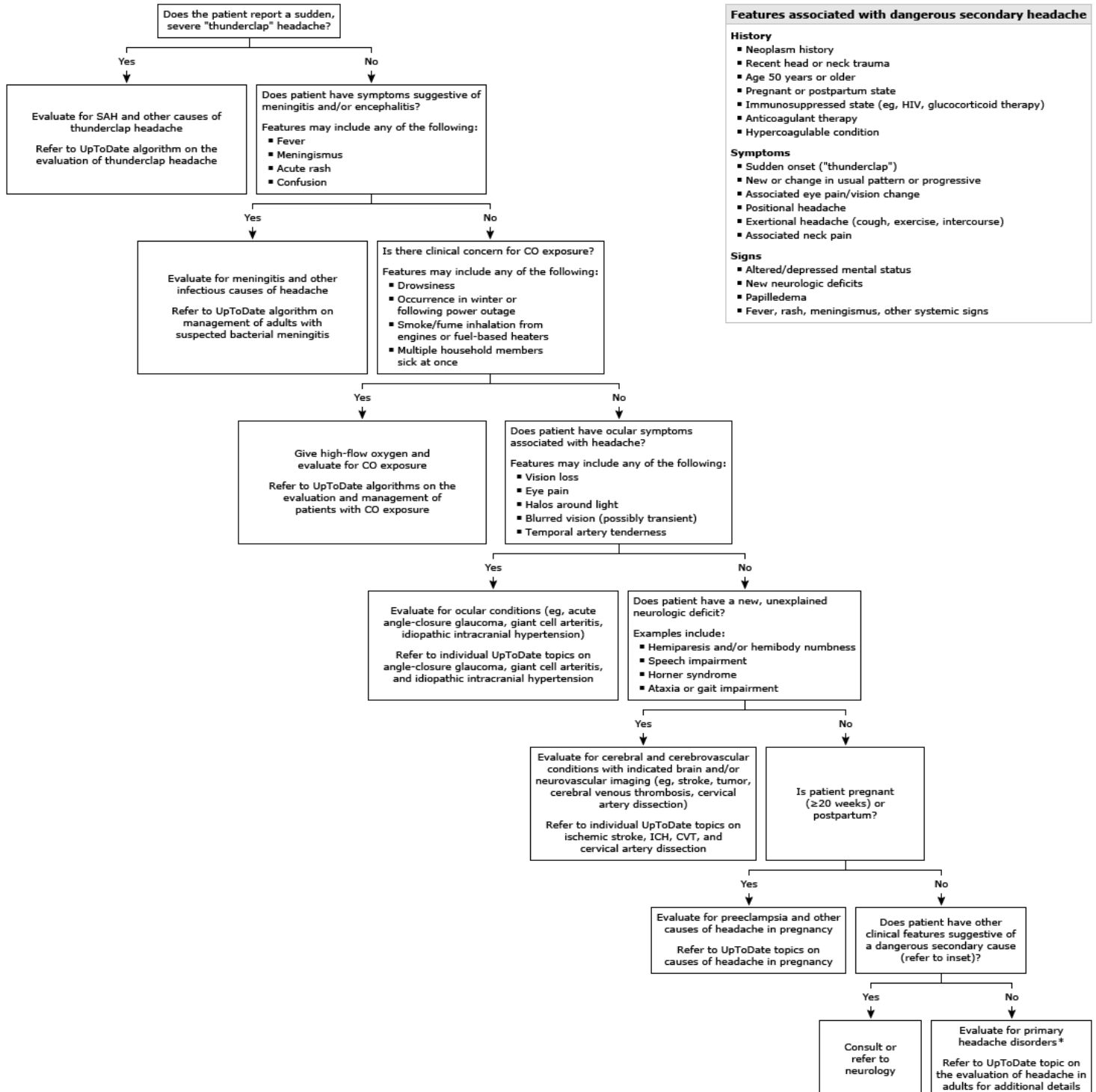
Pertinent Negatives:

- Relatively negative ROS (excepts as noted above in Neuro, GI, HEENT) including absence of ipsilateral autonomic signs (tearing, rhinorrhea) and syncope or loss of consciousness (LOC)
- No exertion associated with onset
- Vital signs (VS) stable, afebrile, mental status intact and age<50

- Negative underlying medical conditions (cancer, immunosuppression), history of headaches or recent infections
- Negative family history of aneurysms, headache syndromes, other neurological problems
- No history of illicit drug use

- Physical exam essentially negative (including no papilledema, nuchal rigidity, facial asymmetry, focal neurologic findings, rash, or temporal pain)
- No environmental toxic exposures
- No history of chronic OTC analgesic use

UptoDate: [Emergency evaluation of the adult with new nontraumatic headache](#)



Assessment Considerations

What two factors in her history may put her at risk of stroke?

- Oral contraceptive pills (OCPs)
- Tobacco use

It is important as physicians that we counsel patients on the risk of stroke associated with OCPs, smoking, and history of headaches.

- Ryan et al. Prevention Opportunities for Oral Contraceptive–Associated Ischemic Stroke *Stroke AHA Journals* 2014;45:893-895 [Link](#) It is important to counsel patients on non-estrogen forms of birth control that the patient could switch to such as Nexplanon, either type of IUD, Depo, or progesterone only OCP).

Labs

What **laboratory tests** are indicated by this patient's history and physical exam?

- A CBC to assist in ruling out infection
- A blood chemistry panel, including electrolytes, glucose, BUN, creatinine

Other lab tests to consider based upon clinical presentation include:

- Lactic acid
- Westergren sedimentation rate (ESR)
- CRP (frequently elevated in temporal arteritis)
- Coagulation studies
- ABG (if indicated)
- Carboxyhemoglobin (to rule out carbon monoxide poisoning)

Lumbar Puncture- Information from-UpToDate

Lumbar punctures (LP) collect cerebrospinal fluid for analysis if there is clinical suspicion of subarachnoid hemorrhage, infectious or inflammatory etiology of headache. The most common use of the LP is to diagnose or exclude meningitis in patients presenting with some combination of fever, altered mental status, headache, or meningeal signs. Examination of the CSF has a high sensitivity and specificity for determining the presence of bacterial and fungal meningitis.

The findings on CSF analysis also may help distinguish bacterial meningitis from viral infections of the central nervous system. However, there is often substantial overlap.

Nonurgent — A nonurgent LP is indicated in the diagnosis of the following conditions. The findings are discussed in the appropriate topic reviews:

- Idiopathic intracranial hypertension (pseudotumor cerebri)
- Carcinomatous meningitis
- Tuberculous meningitis
- Normal pressure hydrocephalus
- CNS syphilis
- CNS vasculitis

Conditions in which LP is rarely diagnostic but still useful include:

- Multiple sclerosis
- Guillain-Barré syndrome
- Paraneoplastic syndromes

LP is also required as a therapeutic or diagnostic maneuver in the following situations:

- Spinal anesthesia
- Intrathecal administration of chemotherapy
- Intrathecal administration of antibiotics
- Injection of contrast media for myelography or for cisternography

CONTRAINDICATIONS — Although there are no absolute contraindications to performing the procedure, caution should be used in patients with:

- Possible raised intracranial pressure
- Thrombocytopenia or other bleeding diathesis (including ongoing anticoagulant therapy)
- Suspected spinal epidural abscess

COMPLICATIONS — LP is a relatively safe procedure, but minor and major complications can occur even when standard infection control measures and good technique are used. These complications include:

- Post-LP headache
- Infection
- Bleeding
- Cerebral herniation
- Minor neurologic symptoms such as radicular pain or numbness
- Late onset of epidermoid tumors of the thecal sac
- Back pain

Radiology

Imaging should be reserved for patients who have red flags or who are not improving with standard medical therapy.

Red Flags include:

- Sudden onset
- Worsening pattern (increasing frequency and/or increasing severity of headaches in an individual with history of headaches)
- Age > 50
- New onset of headaches with an underlying medical condition (e.g. cancer, Lyme disease, HIV)
- Headache with concomitant systemic illness (e.g. fever, neck stiffness, cutaneous rash)
- Focal neurologic signs and/or symptoms
- Papilledema
- Headache secondary to trauma
- Headache during pregnancy and/or postpartum
- Prolonged or bizarre aura
- Triggered by cough, exertion and/or Valsalva

For the head, CT remains the modality of choice over MRI because of speed, cost, and availability. A non-contrast CT is valuable in terms of acute stroke, severe acute headache and coma. MRI is superior to CT in visualizing specific soft tissue structures and vasculature.

Check out the following link:

[American College of Radiology ACR Appropriateness Criteria Headache](#)

Consider the patient in our case?

1. Would you order imaging in this patient's case?
2. What type of imaging would you order?
3. When would you order a head CT vs an MRI?
4. When would you order a head CT with contrast vs without contrast?

Review of the following article from Radiopaedia can be helpful to start answering these questions, [CT head-When to use contrast, when not to?](#)

Use this [Head CT](#) link to view a compilation of images put together by our very own Dr. Smith to review a NORMAL Head CT. **This will not be tested** but is a great way to start to familiarize yourself with imaging. As you get ready to move into 3rd year, this would be a great review resource.

The following video describes how to view a [CT Head from YouTube by Learn Neuroradiology Dr. Brent Weinberg, MD/PhD](#). This is a basic introductory way to approach a head CT.

Plan

Based off this analysis our patient likely had?

Possibly presenting with Migraine w/aura. To have an official diagnosis the individual needs to have had at least **two** attacks including B and C criteria. Our patient has only had this one event.

Migraine without aura
Nausea, vomiting, or both
Photophobia and phonophobia
E. Not better accounted for by another ICHD-3 diagnosis
Migraine with aura
A. At least two attacks fulfilling criteria B and C
B. One or more of the following fully reversible aura symptoms:
Visual
Sensory
Speech and/or language
Motor
Brainstem
Retinal
C. At least three of the following six characteristics:
At least one aura symptom spreads gradually over ≥ 5 minutes
Two or more symptoms occur in succession
Each individual aura symptom lasts 5 to 60 minutes
At least one aura symptom is unilateral
At least one aura symptom is positive [¶]
The aura is accompanied or followed within 60 minutes by headache
D. Not better accounted for by another ICHD-3 diagnosis

With this presentation our patient experienced:

- Visual aura
- Sensory aura
- Language aura

This could possibly fall under the brainstem aura (basilar-type migraine) with tinnitus, numbness, tingling, aphasia

Per UpToDate:

Migraine with aura is a headache disorder characterized by attacks of transient neurologic symptoms usually followed by headaches. Those symptoms last for 2-72 hours. A typical migraine attack progresses through 4 phases: the prodrome, the aura, the headache, and the postdrome.

The aura of migraine usually develops gradually over more than five minutes. Less often, the aura develops more acutely (ie, in less than five minutes). The acute onset of aura makes confusion with a transient ischemic attack (TIA) or stroke more likely. In one case series, four patients (2 percent) had exclusively acute-onset visual aura.

Visual aura — A visual aura classically begins as a small area of visual loss often just lateral to the point of visual fixation. It may either appear as a bright spot or as an area of visual loss. Over the following five minutes to one hour, the visual disturbance often expands to involve a quadrant or hemifield of vision. Along the expanding margin, geometric shapes or zigzagging lines often appear. The shapes account for one of the common names for aura, the "fortification spectrum," because of the resemblance of the aura to the walls of a medieval fortress. The positive visual phenomena may assume a sickle or C-shape, expanding over time toward the peripheral visual field, leaving a scotoma or area of complete visual loss in their wake. As the aura moves off into the peripheral visual field, it often assumes a shimmering or scintillating quality. As the aura resolves, vision usually returns first to the areas of central vision initially affected by the aura.

Sensory aura — The sensory aura is also common and typically follows the visual aura within minutes, although it may also occur without the visual aura. A sensory aura usually begins as a tingling in one limb or on one side of the face. As the tingling sensation migrates across one side of the face or down the limb, numbness is left in its wake that may last up to an hour. The sensory aura may also move inside the mouth, affecting the buccal mucosa and half the tongue. The slow spread of positive symptoms (scintillations or tingling) followed by negative symptoms (scotoma or numbness) is quite characteristic of migraine aura and is not typical for an ischemic event.

Language aura — Less common than the visual and sensory auras is the language or dysphasic aura. Language auras cause transient problems that may run the gamut from mild wording difficulties to frank dysphasia with paraphasic errors.

Motor aura — In the rarest of auras, motor aura, the limbs and possibly the face on one side of the body become weak. Because of information related to the genetic basis of the motor aura, it has been separated from the other forms of aura and classified as **hemiplegic migraine**. The aura symptoms may occur either singly or in sequence, but they do not generally occur simultaneously.

Aura without headache — Some patients may experience aura without an associated headache. Migraine aura without headache (also known as migraine equivalent and acephalgic migraine) manifests as isolated aura unaccompanied by headache. In a Danish case study, 38 percent of patients reported having both attacks of migraine aura without headache as well as migraine aura with headache, and 4 percent had exclusively migraine aura without headache.

Auras without headache are more common in individuals aged >50 years old who have a previous history of migraine with aura and were formerly called late-life migraine accompaniments. The most common symptoms are visual auras, followed by sensory auras (paresthesia), speech disturbances, and motor auras (weakness or paralysis). The most common presentation is a gradual evolution of aura symptoms with spread of transient neurologic deficits over several minutes and serial progression from one symptom to another. These events may be confused with transient ischemic attacks.

Hemiplegic migraine — The primary feature that separates hemiplegic migraine from other types of migraine with aura is the presence of motor weakness as a key manifestation of aura in the attacks. In addition to **weakness**, the aura in hemiplegic migraine also includes visual, sensory, and/or language

disturbance. Symptoms associated with migraine with brainstem aura such as vertigo or diminished consciousness may also occur in some patients. Some patients report prolonged auras lasting several hours. Most patients report headache with or after onset of the aura. Attacks can recur for many years, but overall prognosis is favorable with waning frequency of attacks over time. Hemiplegic migraine may occur as a familial or sporadic form.

See the following link to read more about Migraine [Pathophysiology, clinical manifestations, and diagnosis of migraine in adults](#)

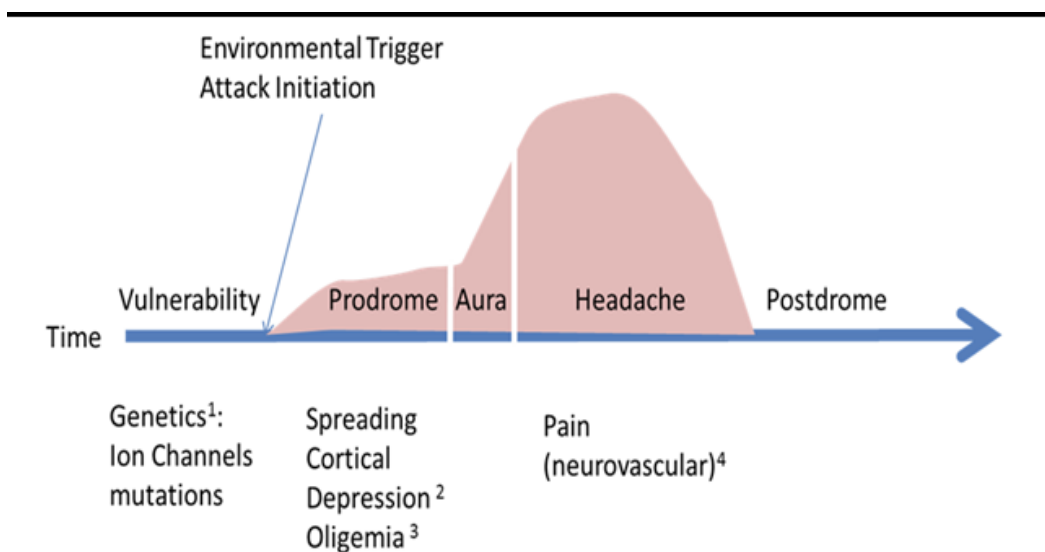
Once emergent situations have been ruled out, we can proceed with management/treatment with may include, avoiding/defining triggers, stress reduction, medications, OMT, and/or lifestyle changes.

Pathophysiology, Classification

Migraine Management

Pathophysiology

Patients with migraine are likely to be vulnerable to outside triggers due to mutations in the regulation of ion balance (e.g. mutations in calcium channels subtypes and Na⁺/K⁺ ATPase subunits) in neurons. The primary neuronal dysfunction leads to a sequence of changes intracranially and extracranially that account for migrainous symptoms, including the three phases of premonitory (prodrome), aura, and headache symptoms.



Blau JN. Migraine: theories of pathogenesis. *Lancet*. 1992;339:1202-1207.

¹Cell. 1996;87:543-552.

Hum Genet. 1995;96(5):604-608.

Neurology. 1998;50:1428-1432.

Nat Genet. 2003;33(2):192-196.

² *J Neurophysiol*. 1944;7:359-390.

Brain. 1994;117 (Pt 1):199-210.

Headache in Clinical Practice. 2nd ed.

England: Martin Dunitz; 2002.

³ *J Neurophysiol*. 1944;7:359-390.

Brain. 1994;117 (Pt 1):199-210.

⁴Goadsby, PJ, Lipton RB, Ferrari, MD. *N Engl J Med* 2002;346:257-270. Copyright (C) 2002.

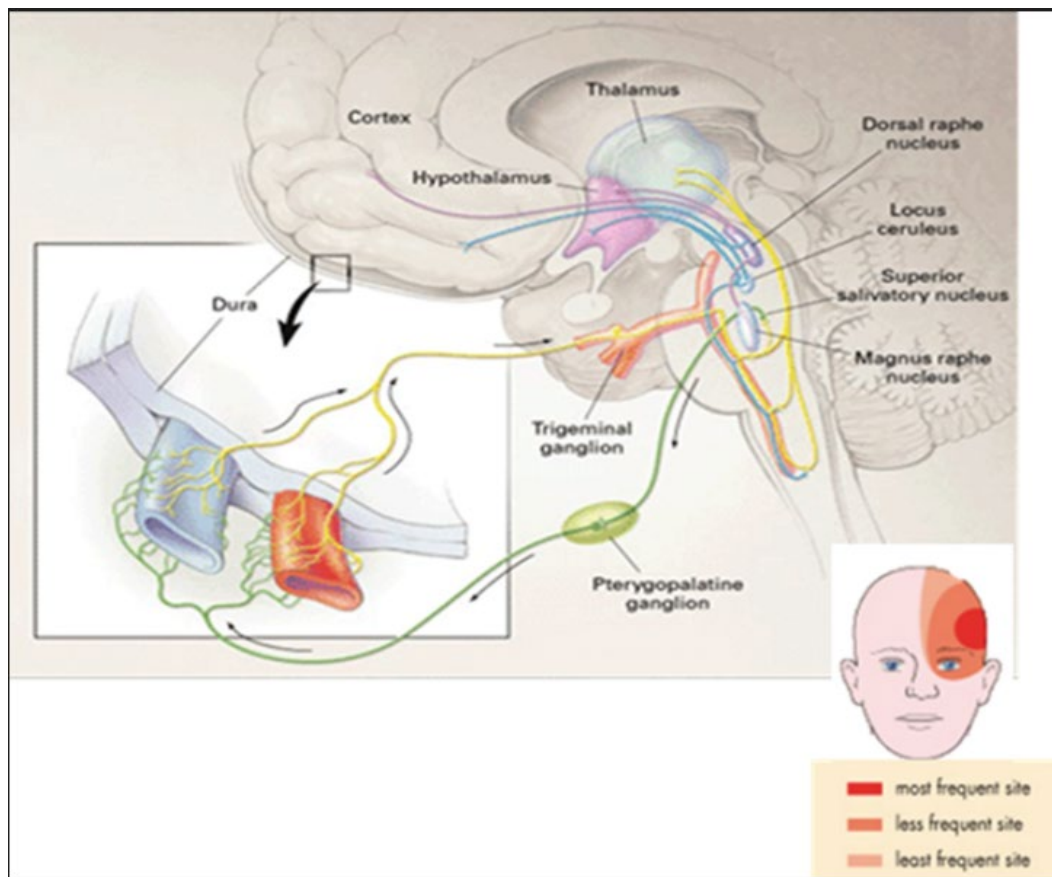
Migraine headaches are typically activated upon environmental triggers that produce changes deep within the brain.

Prodrome: Once initiated a cascade of central cortical depression and reduced blood flow (oligemia) spreads slowly throughout the cortex.

Aura: Changes in nerve cell activity and blood flow, associated with a brief wave of hyperemia followed by prolonged oligemia may (18%-31% of patients with migraine) result in producing visual disturbance, numbness, or tingling, and dizziness depending on which region of the cortex is most affected.

Headache Pain: Pain results from a combination of cranial blood vessel dilation, trigeminal innervation of the blood vessels, and the reflex connections from the trigeminal system with cranial parasympathetic outflow. Central activation of parasympathetic outflow (see green neuron and pterygopalantine ganglion below) produces vascular inflammation in blood vessels of the dura. The inflammation, in turn, stimulates the trigeminal nerve (see yellow and trigeminal ganglion below) that relays nociceptive inputs to the spinal cord resulting in a severe or throbbing pain sensation. In particular, the involvement of the ophthalmic division of the trigeminal nerve and its overlap with layers in the C1 and C2 in the spinal cord produce a type of referred pain that is perceived to be in the frontal and temporal region commonly found in the migrainous headache (see inset). This process can cycle for 4-72 hours.

- **Neurovascular-Reflex key neurotransmitters:** Stimulation of the trigeminal ganglion results in release of vasoactive neuropeptides, including serotonin, dopamine, substance P, calcitonin gene-related peptide, and neurokinin A. Release of these neuropeptides results in the process of neurogenic inflammation. The two main components of this sterile inflammatory process are vasodilation and plasma protein extravasation. Neurogenic inflammation is responsible for the pain of migraine. Elevated levels of vasoactive neuropeptides have been found in the cerebrospinal fluid of patients with chronic migraine, suggesting chronic activation of the trigeminovascular system in these patients. (Neurogenic inflammation may lead to the process of sensitization. Goadsby, PJ, Lipton RB, Ferrari, MD. Migraine. Current Understanding and Treatment. *N Engl J Med* 2002; 346:257-270).
- **Central Neurotransmitter Pathways involved in migraine features:** The symptoms of the migraineur are likely to involve both the serotonergic and dopaminergic pathways in the brain. Peripheral serotonergic pathways that involve the vasculature are important in the cascade of events that produce migraine attacks. In addition, serotonin, released from brainstem nuclei may underlie the migraine symptoms that involve mood and appetite. Data also support a role for dopamine in the pathophysiology of certain subtypes of migraine. For example, migrainous attacks can be induced by dopaminergic stimulation. Moreover, there is dopamine receptor hypersensitivity in migraineurs that may be exhibited via yawning, nausea and vomiting, and hypotension. Dopamine receptor antagonists are effective therapeutic agents in migraine management, especially when given parenterally or concurrently with other antimigraine agents. (*Funct Neurol.* 2000;15 Suppl 3:171-81)



Classification

[International Classification of Headache Disorders, 3rd edition](#)

ICHD-3 criteria for migraine without aura:

Diagnostic criteria:

A. At least five attacks¹ fulfilling criteria B-D

B. Headache attacks lasting 4-72 hr (untreated or unsuccessfully treated)^{2;3}

C. Headache has at least two of the following four characteristics:

1. unilateral location
2. pulsating quality
3. moderate or severe pain intensity
4. aggravation by or causing avoidance of routine physical activity (eg, walking or climbing stairs)

D. During headache at least one of the following:

1. nausea and/or vomiting
2. photophobia and phonophobia

E. Not better accounted for by another ICHD-3 diagnosis.

ICHD-3 criteria for migraine with aura:

A. At least two attacks fulfilling criteria B and C

B. One or more of the following fully reversible aura symptoms:

1. visual
2. sensory
3. speech and/or language
4. motor
5. brainstem

6. retinal

C. At least three of the following six characteristics:

1. at least one aura symptom spreads gradually over ≥ 5 minutes
2. two or more aura symptoms occur in succession
3. each individual aura symptom lasts 5-60 minutes¹
4. at least one aura symptom is unilateral²
5. at least one aura symptom is positive³
6. the aura is accompanied, or followed within 60 minutes, by headache

D. Not better accounted for by another ICHD-3 diagnosis.

ICHD-3 criteria for migraine with typical aura:

A. Attacks fulfilling criteria for 1.2 *Migraine with aura* and criterion B below

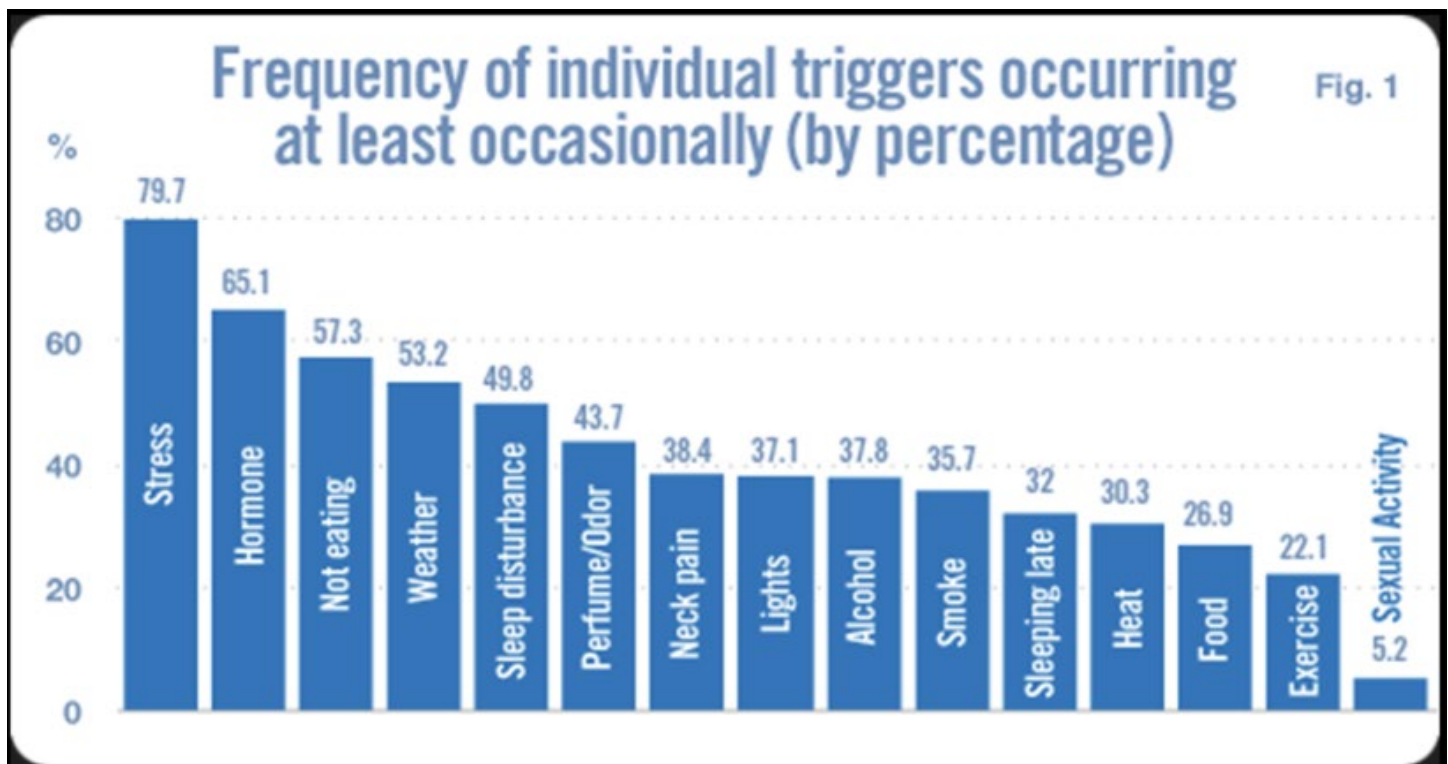
B. Aura with both of the following:

1. fully reversible visual, sensory and/or speech/language symptoms
2. no motor, brainstem or retinal symptoms.

Treatment Options

General Recommendations and Patient Education

1. Identify triggers and avoid triggers as much as possible. Common triggers include:
 1. Changes in daily cycles (e.g. menses, not eating breakfast, working late hours, oversleeping)
 2. Changes in environment (e.g. weather, altitude, bright lights, odors, cigarette smoke)
 3. Diet and medications (e.g. alcoholic beverages, cheeses, chocolate, nitrites)
 4. Mental stress (including anxiety, depression, anger, fear)



[Martin/Kelman]

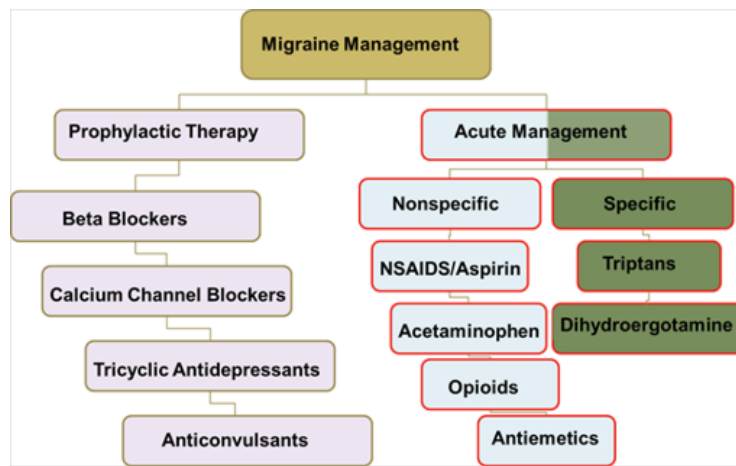
2. Biofeedback, education, and psychological intervention
3. Lifestyle modifications are the cornerstone of prevention: sleep hygiene, stress management, healthy diet, and regular exercise.
4. Pharmacologic Management
 1. Prophylactic medication if attacks are frequent, severely debilitating, or not controlled by acute interventions
 2. Acute interventions- abortive medications
 3. Consider botox injections

Pharmacologic

*Stop OCPs, advise against use in the future-advise on use of no estrogen birth control options

There are two principal pharmacologic directions in the treatment of migraine with medications:

- 1) Acute treatment of pain
- 2) Reduction in the number of attacks with prophylactic treatments. Little is known regarding the mechanism of how prophylactic therapies work.



• **Drug list**

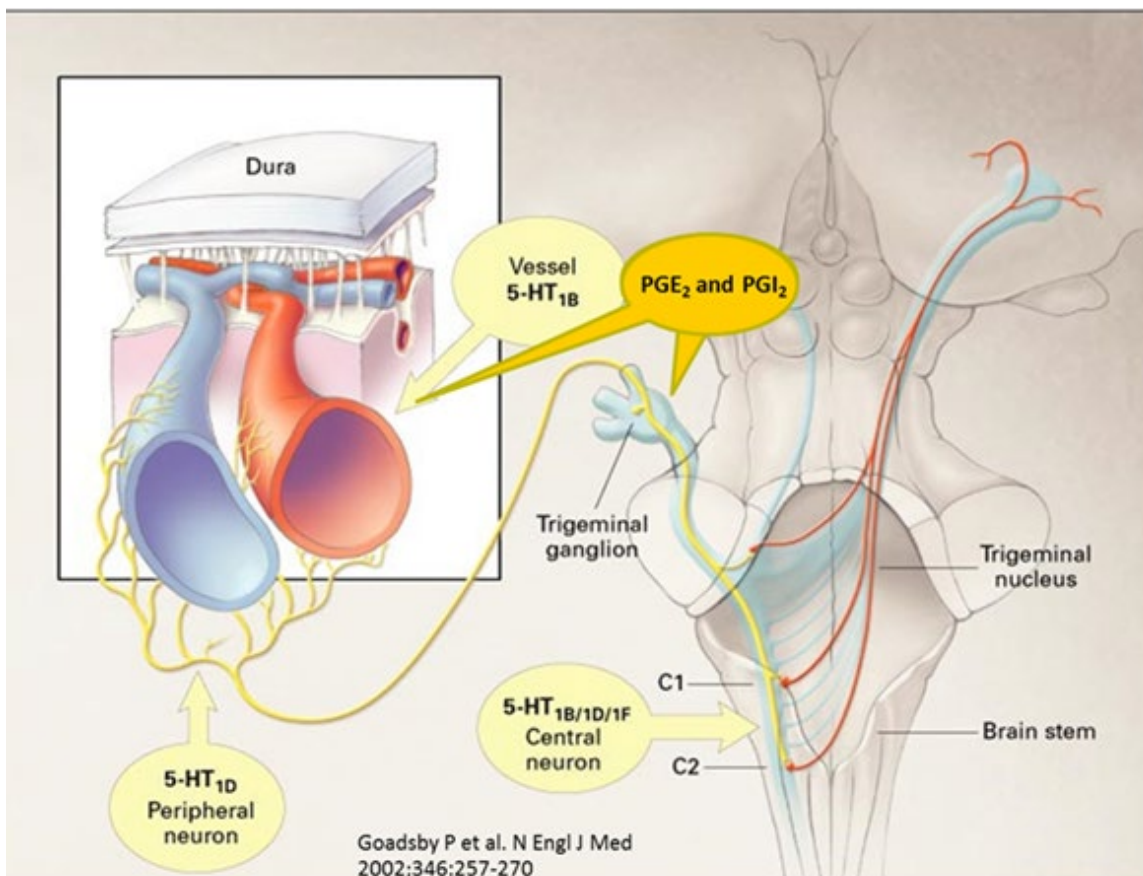
- **Triptans: Migraine Attacks**
 - Sumatriptan
 - Almotriptan
 - Naratriptan
 - Rizatriptan
 - Zolmitriptan
- **NSAIDS**
 - Naproxen, Ibuprofen, Tolfenamic acid
- **Acetaminophen**
- **Opioids**
 - Butorphanol, etc. most covered with opioid lecture
- **Dopamine Antagonists**
 - Metoclopramide
 - Prochlorperazine
 - Chlorpromazine
- **Ergot alkaloids**
 - Dihydroergotamine, Ergotamine, methysergide
- **Beta-blocker**
 - Propranolol
- **Tricyclic antidepressant**
 - Amitriptyline, nortriptyline (covered antidepressant lecture)
- **Anticonvulsants**
 - Topiramate, valproate, gabapentin, lamotrigine (covered in anti-seizure lecture)
- **Calcium channel blocker**
 - verapamil

Acute: Symptomatic

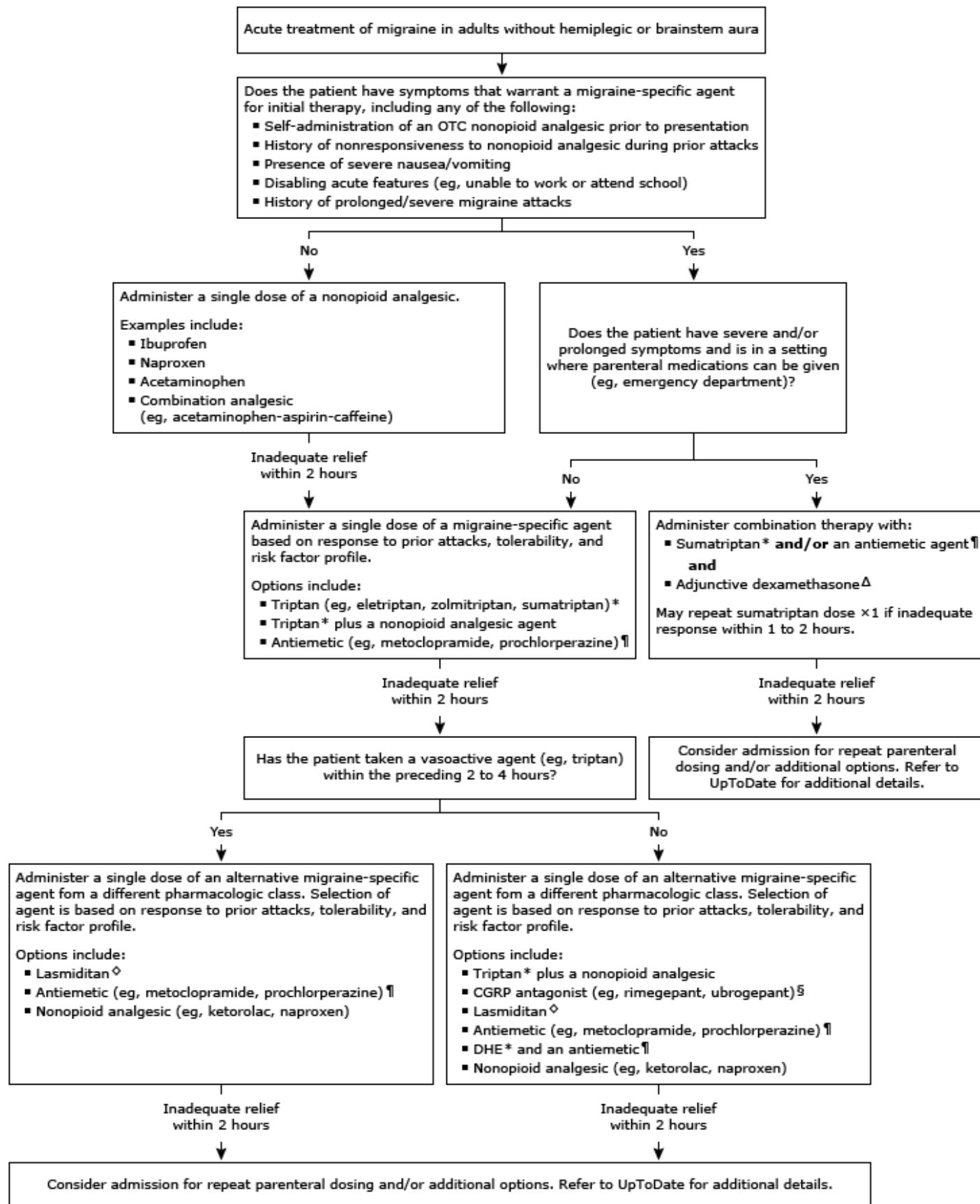
Chronic: Prophylactic

Acute treatment of symptoms --involves a nonspecific decrease in nociception via NSAIDs/Aspirin, acetaminophen and opioids. Of these the safest options are the non-opioid pain relievers. The more specific drugs (a class of drugs called triptans) revolve around suppressing the vascular inflammation and the concomitant activation of Trigeminal neuronal afferents. This later class of drugs represents the most effective pharmacologic treatment option. In addition, if vomiting and nausea are present anti-nausea drugs can be given concomitantly.

The figure below demonstrates the target areas for the most common drugs used to treat acute migraine headaches (NSAIDs and Triptans). NSAIDs and aspirin can suppress the production of vasodilatory prostaglandins PGE₂ and PGI₂ that when made can increase vascular inflammation. In addition, prostaglandins can activate neurons in the trigeminal ganglion. Triptans are a class of drugs that are selective agonist for serotonin 5-HT_{1D} and 5-HT_{1B} receptors. These receptors play two vital roles in the process of migraine headaches. 1) Activation of 5-HT_{1B} receptors produces vasoconstriction thus opposing the vasodilatory effects on dura blood vessels. 2) Activation of 5-HT_{1D} receptors reducing the ability of afferent sensory fibers from firing. This results in less activation of trigeminal ganglion. These receptors play a vital role in the pathophysiology of migraine headaches and why they are considered a specific treatment for this disease.



Per to UpToDate: [Acute treatment of migraine in adults](#) (Those not having brainstem aura* unlike our patient case)



Acute therapy — We suggest an antiemetic medication or a nonsteroidal anti-inflammatory drug (NSAID) for initial acute symptomatic treatment of patients with MBA. Oral NSAIDs are widely available and inexpensive and may be preferred for patients with milder symptoms. Intravenous (IV) or intramuscular (IM) NSAIDs or antiemetic medications may be preferred for patients with moderate to severe headaches or those that are associated with prominent nausea and vomiting.

For patients with mild to moderate MBA symptoms, options include:

- Ibuprofen 400 to 600 mg orally once
- Naproxen sodium 275 to 750 mg orally once
- Diclofenac 50 to 100 mg orally once

For patients with moderate to severe MBA symptoms or significant nausea/vomiting, options include:

- Ketorolac 30 mg IV or 60 mg IM once (lower doses may be warranted for patients age ≥65 years, <50 kg body weight, or those with kidney impairment)
- Prochlorperazine 10 mg IV, IM, or orally once
- Metoclopramide 10 mg IV or orally once

NSAIDs and antiemetic agents may also be used as combination therapy. [Diphenhydramine](#) (25 to 50 mg orally) may be given along with antiemetic medications to prevent akathisia and acute dystonic reactions.

Treatment of Acute Attacks based on American Academy of Neurology Guidelines. [Neuropsychiatr Dis Treat](#). 2013; 9: 709–720. The basis for these recommendations includes the severity of symptoms and the effectiveness of the drug compared to the risk profile of that drug.

- **Mild intermittent Headaches**
 - NSAIDS (common for pain) ± anti-emetics (if nausea and vomiting present)
 - Anti-emetics: Dopamine Antagonists (adjunctive therapy)
 - Metoclopramide
 - Prochlorperazine
 - Chlorpromazine
 - Selective Therapies if NSAIDs fail:
 - Serotonin agonists (triptans)
 - Most are oral but for rapid relief or rescue therapy there are SC and nasal sprays available
 - **Moderate-to-severe**
 - Triptans
 - Combinations: NSAID with triptan e.g. Sumatriptan 85mg-Naproxen 500mg (Treximet)
 - Opioid analgesics
 - Parenteral narcotics may be useful for infrequent, severe headaches unresponsive to other pharmacologic approaches
 - Nonselective serotonin agonists still available—less common (Ergot Derivatives)

- Ergotamine
- Dihydroergotamine

AAN: Evidence Based Guidelines—US Headache Consortium recommendations:

Preventative Therapy: Roughly about 10% of Americans who suffer from migraines are treated with daily medications to reduce the number of attacks. Approximately, 40% of patients who suffer from migraine headaches would benefit from daily treatment. [Neuropsychiatr Dis Treat](#). 2013; 9: 709–720

Preventative therapy may be more appropriately guided by the following.

1. Recurring migraines that, in the patient's' opinion, significantly interfere with their daily routines, despite acute treatment
2. Frequent headaches
3. Contraindication to, failure of, or overuse of acute therapies
4. Adverse events with acute therapies
5. The cost of both acute and preventative therapies
6. Patient preference
7. The presence of uncommon migraine conditions, including hemiplegic migraine, basilar migraine, migraine with prolonged aura, or migrainous infarction (to prevent neurologic damage)

Via a variety of mechanism all of these drugs inhibit the spreading of cortical depression.

-

Osteopathic Considerations (ABCS)

What are *relative* contraindications to treating patients with osteopathic cranial manipulative medicine?

- Acute or recent CVA, TIA, or hemorrhage
- Acute or unstable head trauma
- Acute skull fracture
- Acute concussion (mTBI)

Remember that we treat *people* in osteopathic medicine, not clinical conditions. Therefore, there is no specific technique for a specific diagnosis including headache. Each patient, regardless of the clinical diagnosis, requires a structural diagnosis within the context of their clinical history to determine what treatment best suits their clinical presentation. There is no algorithm for the osteopathic treatment of patients-no one person will have the same constellation of musculoskeletal abnormalities. We can, however, consider each patient in the context of the A (Autonomics), B (Biomechanics), C (Circulation), S (Screening) to begin to discover where we may want to aim our diagnosis and treatment.

Further reading: FOM 4th edition Chapter 43D: Osteopathic Considerations in Neurology pg 1258

- **A (Autonomics)**

- ***Sympathetics***

- Originate at T1-T4- travel to the superior cervical ganglia which lies anterior to the C2 transverse process. Sympathetic fibers then travel along the internal carotid, the external carotid and cranial nerves II, III, IV, VI, and IX

- ***Parasympathetics***

- Begin at 4 nuclei in the brainstem and are associated with CN III, VII, IX, X
 - They synapse in a ganglia- either the ciliary, sphenopalatine, submaxillary, or otic
 - “When evaluating patients with migraine, pay attention to the cranium, in particular the sphenobasilar area, the temporal area, the sphenosquamous pivot, and the jugular foramen. Nausea and vomiting with migraine may be due to vagal involvement at the jugular foramen accompanied by tension on the dural sleeve. Also consider the suboccipital region, C2 (vagus involvement), and the upper thoracic spine and the associated ribs” [Richards].

- **B (Biomechanics)**

- ***Head-*** remember to evaluate for static and dynamic findings as well as the motion of the SBS (sphenobasilar synchondrosis), rate of the CRI, evaluation of the TMJ, temporalis, masseter, and pterygoids (as bruxism and TMJ syndrome/TMD can contribute to headaches)
 - ***Cervical Spine-*** cervical spine mechanics, fascia, and muscles may have an effect on the greater and lesser occipital nerve or refer pain themselves (sclerotomal or referral pain patterns)
 - ***Thoracic Spine-*** as it connects through fascial and muscular connections and importance of addressing sympathetic tone (as above)
 - ***Sacrum/Pelvis-*** don't forget the link between the cranium and the sacrum through the dural attachments (reciprocal tension membrane). The sacrum should be able to move with respiration (breathing)-flexion and extension, as well as with the CRI with cranial counterrotation and nutation.

- **C (Circulation)**

- ***Tentorium Cerebelli-*** dural membrane that is considered a diaphragm and if restricted may impede CSF flow
 - ***Thoracic inlet-*** lymphatic and vascular drainage from the head may be impeded here
 - ***Diaphragm-*** major lymph/venous pump

- **S (Screening)**

- ***Postural issues-*** upper crossed syndrome (to be reviewed later this semester), forward head/neck posture, scoliosis. Remember that abnormal tensions through muscles and fascia distant from the head can still have an effect.
 - ***Leg length discrepancy-*** or sacral base unleveling can cause compensatory postures all the way up

Mind/Body/Spirit considerations- life stressors can contribute to increases in many types of common headaches- increased cortisol and increased sympathetic tone lead to an array of physiologic symptoms

Osteopathic Considerations Treatment

OMT Techniques for Headaches*

***Please note that although there is not a specific HA COAR lab this year, these techniques are covered in other labs. These techniques listed here are a review and just a reminder of some of the many OMT techniques you have learned already that could help patients with headaches.**

TMJD considerations: Assessing for temporomandibular joint dysfunction (TMD) should be done in anyone with chronic headaches. Assessment should include:

- Observing jaw opening and closing
- Palpation of temporalis, buccinator, and medial/lateral pterygoids
- Cranial strain pattern at the SBS (remember the attachments of the pterygoids at the pterygoid plates of the sphenoid) and motion of the mandible and temporal bone (internal rotation or external rotation)
- OA, AA, and upper cervical spine
- Innervation of the TMJ is from the mandibular branch of the trigeminal nerve.

Occipitomastoid (OM) Suture Release may assist with venous drainage from the cranium as well as assist in restoring motion to the occiput and temporal bone. Recall that CSF drains through arachnoid granulations into the venous sinuses. Where does it go from there? The venous sinuses ultimately drain from the sigmoid sinus into the internal jugular vein which passes through the jugular foramen. The jugular foramen is anteromedial to the OM suture.

Keys to helping a patient with a headache with OMT:

- Make an accurate clinical diagnosis
 - Primary Headache (migraine, rebound, cervicogenic, etc) vs.
 - Secondary Headache (systemic infection, intracranial hemorrhage, tumor, etc)
- Make an accurate structural diagnosis
 - Treat what you find!
 - Re-evaluate both your clinical and structural diagnosis

Key things to consider when treating with OMT for a patient with a migraine:

- Free the sphenobasilar and temporal regions.
- Treat the cervical dysfunctions with indirect procedures, balanced ligamentous tension, or very gentle direct myofascial release with care to not over stimulate the vagus nerve.
- Remember: during a migraine, your patient will NOT want you to rotate their head and neck around. For this reason, HVLA to the cervical region is not recommended during the acute phase of a migraine.
- Use any procedure to normalize the function of the thoracic and rib dysfunctions.
- Treat counterstrain points in the suboccipital area and the inion point

Osteopathic Research

(Note- no test or quiz questions will come from this research section- but it's important for you to know about what research is out there! Do a literature search- what research on OMT and headaches can you find?)

- Cerritelli, F. et al. Osteopathy for primary headache patients: a systematic review. *J Pain Res* 2017;10:601-611
 - <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5359118/>
 - The results from this systematic review **show a preliminary low level of evidence that OMT is effective in the management of headache**. However, studies with more rigorous designs and methodology are needed to strengthen this evidence.
- Voigt K, Liebnitzky J, Burmeister U et al. Efficacy of osteopathic manipulative treatment of female patients with migraine: results of a randomized controlled trial. *J Altern Complement Med*. 2011;17(3):225-230.
 - <https://www.ncbi.nlm.nih.gov/pubmed/21385086>
 - This study affirms the effects of OMT on **migraine headache in regard to decreased pain intensity and the reduction of number of days with migraine as well as working disability**, and partly on improvement of HRQoL. Future studies with a larger sample size should reproduce the results with a control group receiving placebo treatment in a long-term follow-up.
- Garcia J, Arnold S, Tetley K, Voight K, Frank RA. Mobilization and manipulation of the cervical spine in patients with cervicogenic headache: any scientific evidence? *Front Neurol*. 2016;7:40. doi:10.3389/fneur.2016.00040
 - <https://www.ncbi.nlm.nih.gov/pubmed/27047446>
 - Seven of the 10 studies had statistically significant findings that subjects who received mobilization or manipulation interventions experienced improved outcomes or reported fewer symptoms than control subjects. These results suggest that **mobilization or manipulation of the cervical spine may be beneficial for individuals who suffer from cervicogenic headache**, although heterogeneity of the studies makes it difficult to generalize the findings.
- Schabert, E. et al Impact of osteopathic manipulative treatment on cost of care for patients with migraine headache: a retrospective review of patient records. *JAOA*. 2009;8(109):403-407
 - [Article on https://ostemed-dr.contentdm.oclc.org](https://ostemed-dr.contentdm.oclc.org)
 - The inclusion of **OMT in a treatment regimen for patients with migraine HA may lower the cost of the treatment regimen**. However, further study is needed to confirm these results.
- Rolle, Guido, Tremolizzo, Lucio, Somalvico, Francesco, Ferrarese, Carlo and Bressan, Livio C.. "Pilot Trial of Osteopathic Manipulative Therapy for Patients With Frequent Episodic Tension-Type Headache" *Journal of Osteopathic Medicine*, vol. 114, no. 9, 2014, pp. 678-685. <https://doi.org/10.7556/jaoa.2014.136>
- Alexander, Joshua. "Resolution of New Daily Persistent Headache After Osteopathic Manipulative Treatment" *Journal of Osteopathic Medicine*, vol. 116, no. 3, 2016, pp. 182-185. [Link](#)

- [Link](#)
 - New daily persistent headache (NDPH) is a rare, poorly understood, treatment-refractory primary headache disorder with heterogeneous symptoms. This case report describes a patient with NDPH that resolved after OMT.
- Lee, Elaine, Moloney, Sean, Talsma, Joel and Pierce-Talsma, Stacey. "Osteopathic Manipulative Treatment Considerations in Tension-Type Headache" *Journal of Osteopathic Medicine*, vol. 119, no. 10, 2019, pp. e40-e41. [Link](#)
- Jara Silva CE, Joseph AM, Khatib M, Knafo J, Karas M, Krupa K, Rivera B, Macia A, Madhu B, McMillan M, Burtch J, Quinonez J, Albert T, Khanna D. Osteopathic Manipulative Treatment and the Management of Headaches: A Scoping Review. *Cureus*. 2022 Aug 9;14(8):e27830. doi: 10.7759/cureus.27830. PMID: 36110479; PMCID: PMC9462953.
 - [Link](#)
 - This review study demonstrates the benefits OMT has on decreasing headache frequency, intensity, and duration in TTH and migraines. OMT has shown to be beneficial, especially for patients seeking alternative non-pharmaceutical and non-invasive treatments. Further studies are needed to evaluate the effects of different OMT techniques, and different combinations of treatments, on other types of headaches.
- Whalen J, Yao S, Leder A. A Short Review of the Treatment of Headaches Using Osteopathic Manipulative Treatment. *Curr Pain Headache Rep*. 2018 Oct 5;22(12):82. doi: 10.1007/s11916-018-0736-y. PMID: 30291550.
 - [Link](#)
 - OMT is a useful non-invasive treatment option with little to no side effects. There are multiple headache types. Migraine, tension-type headache, combat-related events, post-traumatic headache, sinusitis, tooth extraction, concussions, and others have all shown benefit from OMT.

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